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Oral Microbiota Composition Predicts Early Childhood Caries Onset

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Abstract

As the most common chronic disease in preschool children in the United States, early childhood caries (ECC) has a profound impact on a child's quality of life, represents a tremendous human and economic burden to society, and disproportionately affects those living in poverty. Caries risk assessment (CRA) is a critical component of ECC management, yet the accuracy, consistency, reproducibility, and longitudinal validation of the available risk assessment techniques are lacking. Molecular and microbial biomarkers represent a potential source for accurate and reliable dental caries risk and onset. Next-generation nucleotide-sequencing technology has made it feasible to profile the composition of the oral microbiota. In the present study, 16S ribosomal RNA (rRNA) gene sequencing was applied to saliva samples that were collected at 6-mo intervals for 24 mo from a subset of 56 initially caries-free children from an ongoing cohort of 189 children, aged 1 to 3 y, over the 2-y study period; 36 children developed ECC and 20 remained caries free. Analyses from machine learning models of microbiota composition, across the study period, distinguished between affected and nonaffected groups at the time of their initial study visits with an area under the receiver operating characteristic curve (AUC) of 0.71 and discriminated ECC-converted from healthy controls at the visit immediately preceding ECC diagnosis with an AUC of 0.89, as assessed by nested cross-validation. *Rothia mucilaginosa, Streptococcus* sp., and *Veillonella parvula* were selected as important discriminatory features in all models and represent biomarkers of risk for ECC onset. These findings indicate that oral microbiota as profiled by high-throughput 16S rRNA gene sequencing is predictive of ECC onset.

Keywords: dental caries, risk assessment, receiver operating characteristic curve, machine learning, biomarkers, 16S rRNA

Introduction

Early childhood caries (ECC) is a multifactorial, chronic oral disease characterized by localized, irreversible destruction of susceptible dental hard tissue in children prior to 6 y of age. It is the most common chronic disease in children in the United States and disproportionately affects those living in poverty, regardless of race, ethnicity, or culture (Dye et al. 2007). Recent national survey data indicate that the overall prevalence of ECC in US preschool children is 24%, ranging from 11% to 72% depending on household income, ethnicity, and other demographic factors (Schroth et al. 2005; Ismail et al. 2008; Kopycka-Kedzierawski et al. 2008; Phipps et al. 2012; Dye et al. 2015). ECC is difficult to treat effectively, and it profoundly affects quality of life. Accordingly, ECC constitutes a tremendous cost to society and imposes an exceptional burden on the most vulnerable children and their families (Casamassimo et al. 2009).

Caries risk assessment (CRA) is a critical component of dental caries management, whereby clinicians assess the likelihood of future dental caries onset to inform preventive measures (Twetman and Fontana 2009). A broad range of environmental, behavioral, and biological factors are thought to influence ECC risk in young children, including diet, saliva composition, dental caries history, genetics, and socioeconomic status (Mejare et al. 2014; Fontana 2015). Combinations of these factors have been incorporated into numerous tools for CRA in common practice today, primarily in the form of standardized risk assessment questionnaires and guidelines. However, the accuracy, precision, and longitudinal validation of these techniques are lacking, especially in preschool age

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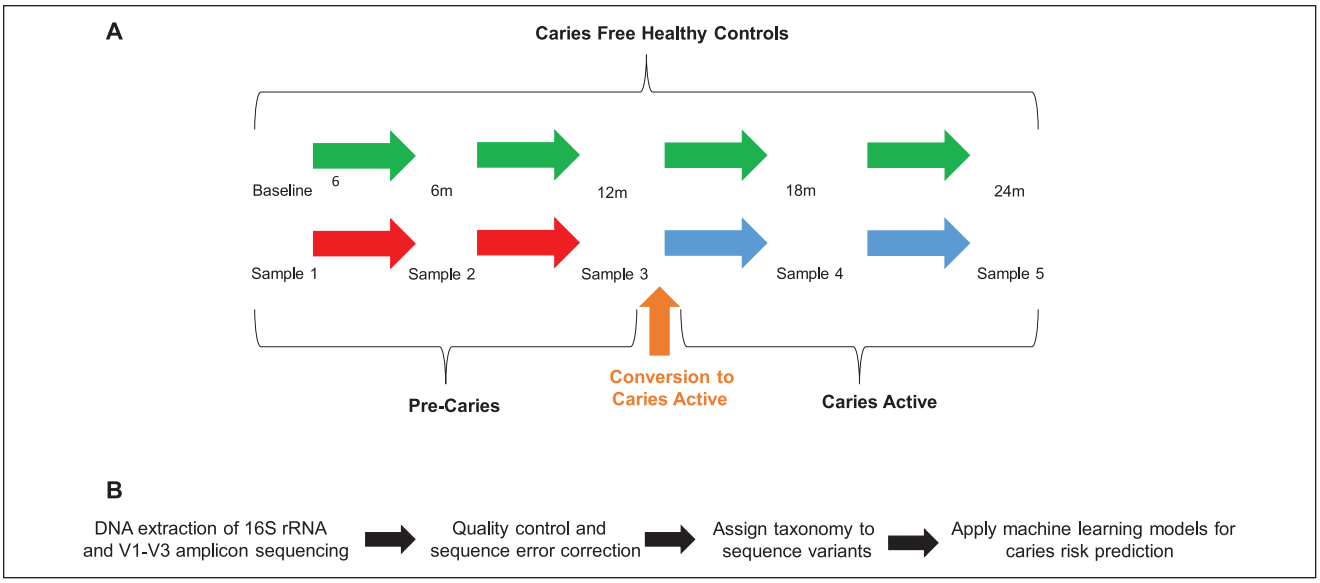


Figure 1. Study overview. (**A**) Sampling scheme overview. Saliva samples were collected at enrollment and every 6 mo thereafter for the duration of the 24-mo study period. Samples were grouped into precaries, caries active, and caries-free healthy controls. (**B**) Summary of analysis pipeline. DNA was extracted from saliva samples, and the V1 to V3 hypervariable region of the 16S ribosomal RNA gene was amplified and sequenced to profile the composition of the oral microbiota. After quality control and taxonomic assignments, profiles of microbiota composition were used to train machine learning models to predict caries risk.

children, and none are considered adequate methods of actionable caries risk assessment (Tellez et al. 2013; Fontana 2015).

Contemporary advances in laboratory technologies have led to both an increased understanding of the microbial contributions to the etiology and pathogenesis of ECC and a recognition of the differences in oral microbiota composition between caries-free and caries-active children (Luo et al. 2012; Jiang et al. 2016; Hajishengallis et al. 2017). In addition to performing metabolic activity that is directly cariogenic, the oral microbiome reflects conditions in the oral cavity arising from other cariogenic factors, such as diet, oral hygiene, and the chemical characteristics of saliva, suggesting its potential utility as a predictor of ECC (Grassl et al. 2016; Hemadi et al. 2017; Kato et al. 2017; Willis et al. 2018). Consequently, 2 recent studies described models of the differences in oral microbiota composition between caries-free and caries-active children used to classify caries-free individuals with more "caries-like" microbiota as at risk for ECC (Teng et al. 2015; Xu et al. 2018). In both studies, the genera *Streptococcus* and *Prevotella* were found to be most discriminatory.

A limitation of cross-sectional evidence is that it cannot differentiate oral microbiota composition that precedes and may be a cause of the disease state from oral microbiota composition that may result from disease onset and is therefore an effect. Accordingly, in the present study, we use longitudinal samples from 56 initially caries-free children aged 1 to 3 y to identify oral microbiota composition that differentiated those children who remained caries free from those who exhibited caries onset within a 2-y follow-up period. We used 2 common machine learning methods, which provide an unbiased and reproducible approach to differentiating children who remained caries free from those who experienced caries onset. In contrast to other recent work, we sought to directly model the signatures of caries risk, rather than simply the extent to which a caries-free sample resembled a caries-active state (Teng et al. 2015; Xu et al. 2018). This approach, combined with our prospective, longitudinal study design following initially caries-free children for 2 y, provides the strongest test to date of the hypothesis that oral microbiota may play a causal role in ECC onset.

Materials and Methods

Detailed materials and methods, including a description of the study population, 16S ribosomal RNA (rRNA) gene sequencing, bioinformatics, and statistical methods of the study are described in the Appendix. Written informed consent was obtained from parent or guardian of all participating infants. The institutional review board at the University of Rochester School of Medicine and Strong Memorial Hospital approved the study. All 16S rRNA sequence reads and generated data sets are publicly available through NCBI accession number PRJNA622300.

Results

Overview of Subjects and Samples

Longitudinal saliva samples were collected from a subset of 56 initially caries-free children, 1 to 3 y of age, to characterize the differences in oral microbiota composition among children with ECC onset within 2 y ("pre-caries," $n = 36$) and those who remained caries free for the 2-y study period ("caries-free healthy controls," $n = 20$) and to predict caries risk (Appendix Table 1 and Fig. 1). As shown in Figure 1, children who remained caries free versus the children who converted to caries-active state were

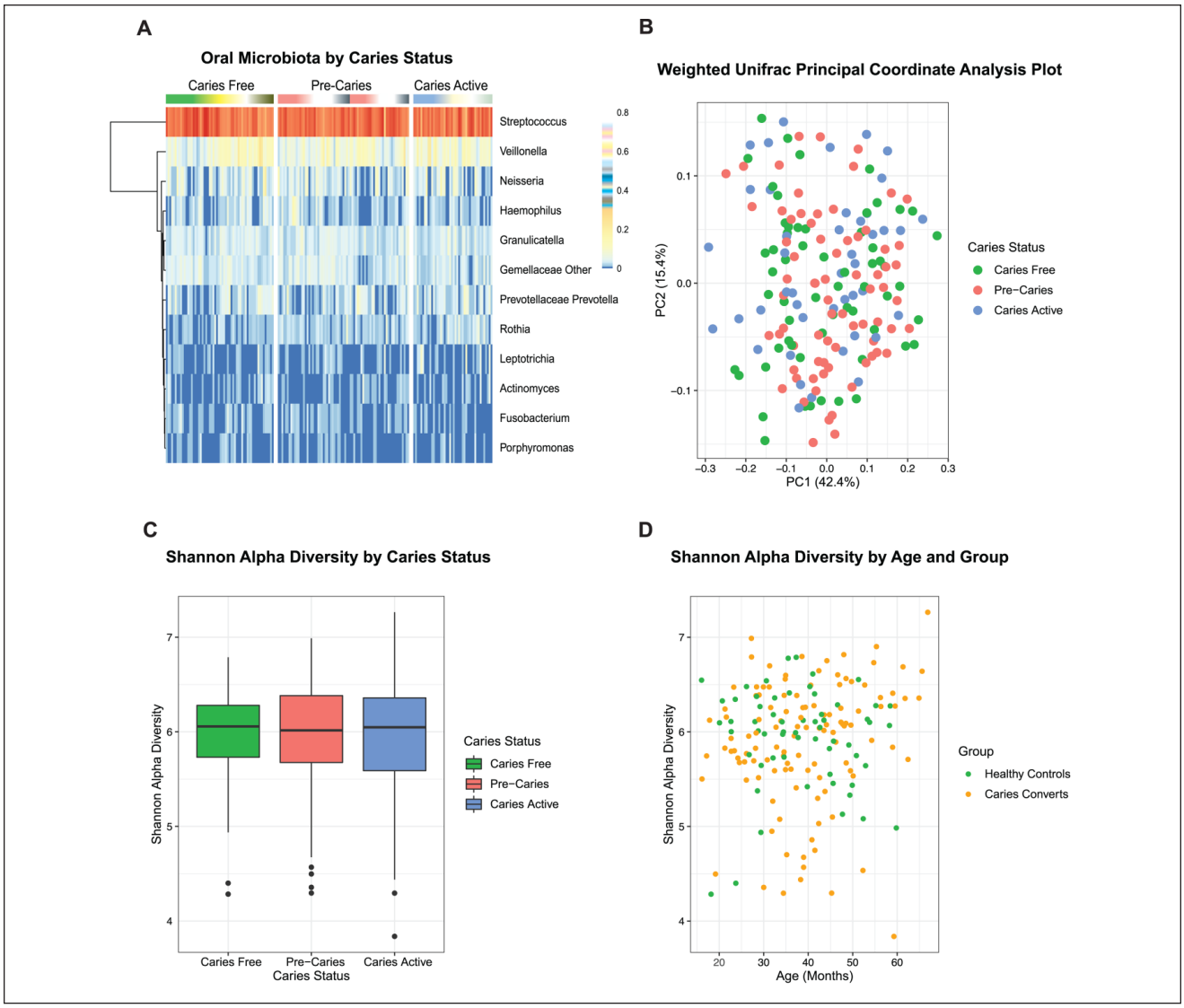


Figure 2. Sample composition and diversity. (**A**) Heatmap showing the relative abundance of the 12 most abundant genre across all samples. Individual samples (thin columns) are grouped based on caries status (broad columns). The genre shown account for 94.2% of the total overall composition of all samples. (**B**) Principal coordinate analysis (PCoA) plot based on the weighted Unifrac distance metric, with samples colored by caries status. The weighted Unifrac distance is a metric of the dissimilarity between the composition of 2 microbiota samples. The PCoA plot is a summary representation of the overall structure of the data and the similarity/dissimilarity relationships between samples. There is no significant clustering or separation based on caries status ($P = 0.18$, PERMANOVA). (**C**) Boxplots of the α diversity of the samples in each caries status group, showing no significant differences ($P = 0.98$, Kruskal-Wallis). (**D**) The α diversity of all samples plotted over age, revealing no significant relationship ($P = 0.18$, Spearman correlation). Samples are colored by subject group: those who developed caries during the study period (caries converts) and those who did not (healthy controls).

similar in age, race, ethnicity, and *Candida* spp. status at baseline. Girls were overrepresented in the healthy group ($P = 0.04$).

Saliva microbiota composition was determined by 16S rRNA gene sequencing. After sample preparation, sequencing, and data quality filtering, 60 samples from caries-free healthy controls and 117 samples from children with ECC onset during the study period were available for analysis. Of the latter, 73 represented a pre-caries state and 44 represented a caries-active state. The minimum number of reads across all samples was 11,671, with an average of 39,751 per sample, representing 4,060 unique 16S rRNA gene amplicon sequence variants, 113 distinct species-level taxonomic assignments, and 85 genera. The top 12 most abundant genera accounted for 94.2% of the overall composition across all samples, with *Streptococcus* being by far the most abundant (61.7% overall abundance), followed by *Veillonella* (9.2%), *Neisseria* (4.1%), *Granulicatella* (3.7%), and an unnamed genus of *Gemellaceae* (3.7%), which were the only genera comprising more than 3% of the overall composition (Fig. 2A). No significant differences existed based on caries status in terms of weighted Unifrac β ($P = 0.18$, PERMANOVA; Fig. 2B) or Shannon and Faith's α ($P = 0.98$ and 0.71, respectively, Kruskal-Wallis; Fig. 2C) diversity, and

α diversity did not exhibit a significant correlation with age ($P = 0.18$ for Shannon and 0.14 for Faith's, Spearman correlation; Fig. 2D).

Caries Status and Microbiota Composition

To identify potential biomarkers of caries risk at the earliest possible time point, we performed univariate tests of each species-level taxonomic unit for differential abundance between pre-caries subjects and healthy control subjects at the time of enrollment. While no taxa were significantly differentially abundant after multiple test correction, a dozen yielded unadjusted *P* values below 0.05 and are visualized here for reference (Fig. 3A). While these marginally significant differences between all pre-caries samples at the time of enrollment based on a simple univariate test are uncompelling on their own, these observations prompted us to examine microbial abundances across caries states in a more rigorous way, using a mixed-effects model to account for repeated sampling of subjects, differences in age, and technical differences arising from differing numbers of sequencing reads per sample. Because ECC onset is thought to be the result of a progressive process, with increasingly cariogenic microbiota developing from healthy to pre-caries to caries-active states, we included all 3 of these states in this model and used the healthy caries-free state as the baseline point of comparison. Under this model, we identified 8 species-level taxa that were significantly differentially abundant between states at a false discovery rate of 0.1 (Fig. 3B). Relative to healthy controls, *Peptostreptococcus* sp. was diminished in the caries-active state (adjusted $P = 0.006$), *Veillonella parvula* was diminished in precaries and caries-active states (adjusted $P = 0.018$ and 0.040, respectively), *Kingella* sp. was enriched in the caries-active state (adjusted $P = 0.018$), *Oribacterium* sp. was diminished in the precaries state (adjusted $P = 0.074$), unclassified *Pseudomonas* was diminished in precaries and caries-active states (adjusted $P = 0.087$ and 0.075, respectively), *Corynebacterium* sp. was enriched in precaries and caries-active states (both adjusted $P = 0.087$), *Selenomonas* sp. was enriched in precaries (adjusted $P = 0.087$), and *Rothia mucilaginosa* was enriched in the caries-active state (adjusted $P = 0.087$). These results confirm the same patterns of enrichment between precaries and healthy states for *Corynebacterium* sp., *Selenomonas* sp., unclassified *Pseudomonas*, and *V. parvula* as were detected in both the baseline-visit and all time point univariate tests. Interestingly, of these, *Corynebacterium* sp., *Selenomonas* sp., *Kingella* sp., unclassified *Pseudomonas, R. mucilaginosa*, and *V. parvula* were also marginally significant at the enrollment time point, suggesting that they may be early and persistent biomarkers of an oral environment developing toward a cariogenic state.

Predicting Caries Risk

Having characterized the differences in microbiota composition between children in a precaries state and those in a healthy caries-free state at the level of individual taxa, we sought to develop models capable of discriminating between the 2 states based on overall microbiota composition, including all taxa as joint predictors. We selected 2 types of common machine learning models that are capable of capturing interactions between features and nonlinear relationships: random forest and gradient boosting classifiers. Given that these models would have the most practical utility if they required only 1 observation per subject, we opted to train and test them on individual time points independently: 1 set of models for the baseline visit and another set of models for the last visit prior to caries diagnosis (plus a set of corresponding samples from healthy control subjects). To get an unbiased estimate of the generalizability of our approach, we adopted an 8-fold nested cross-validation testing and training regime. This involved, for each type of model and each time point, randomly partitioning the samples into 8 equally sized groups, training a model on 7 of the groups using cross-validation to tune the hyperparameters, and then testing the model on the remaining group that was held out from training. This was repeated 8 times, yielding 8 independent models, each tested on a different group of samples that were held out from training. The resulting metrics of model performance and feature importance were computed by combining or averaging all 8 models (Allen 1974).

Both models based on baseline visit samples exhibited significant predictive power, distinguishing between those children with ECC onset during the following 24 mo and those who remained caries-free. For predictions based on the enrollment visit, the random forest classifier performed best, achieving an area under the receiver operating characteristic (ROC) curve (AUC) of 0.71—which is better than random chance at a significance of $P = 0.005$—and an accuracy of 73.2% with a sensitivity of 0.806 and a specificity of 0.600 at the optimum threshold for discrimination (Fig. 4A). The gradient boosting classifier performed slightly worse, achieving an AUC of 0.65 ($P = 0.033$) and an accuracy of 71.4% at the optimal threshold (sensitivity = 0.861 and specificity = 0.450) (Fig. 4B). The models based on the last visit prior to caries diagnosis both performed better than the baseline visit models (test for AUC improvement: random forest $P = 0.020$, gradient boosting $P = 0.012$), with the random forest again outperforming the gradient boosting classifier, achieving an AUC of 0.89 ($P < 0.001$) with an accuracy of 85.5% at the optimal threshold (sensitivity = 0.857 and specificity = 0.850) (Fig. 4C). The AUC of the gradient boosting classifier was 0.86 ($P < 0.001$) with an accuracy of 83.6% at the optimal threshold (sensitivity = 0.771 and specificity = 0.950) (Fig. 4D). Notably, the marked improvement in performance of both models based on the last precaries visit over those based on the enrollment visit was due almost entirely to greatly improved specificity due to a reduction in false positives.

Each type of classification model has a corresponding metric of feature importance, roughly interpretable as the extent to which inclusion of the feature in the model improves the accuracy of the model's predictions. The topmost important features cumulatively accounting for at least 50% of the total feature importance in a model were selected as "important" for that model. For the baseline visit classifier and the immediate pre-caries classifier, the random forest had 14 and 16 features

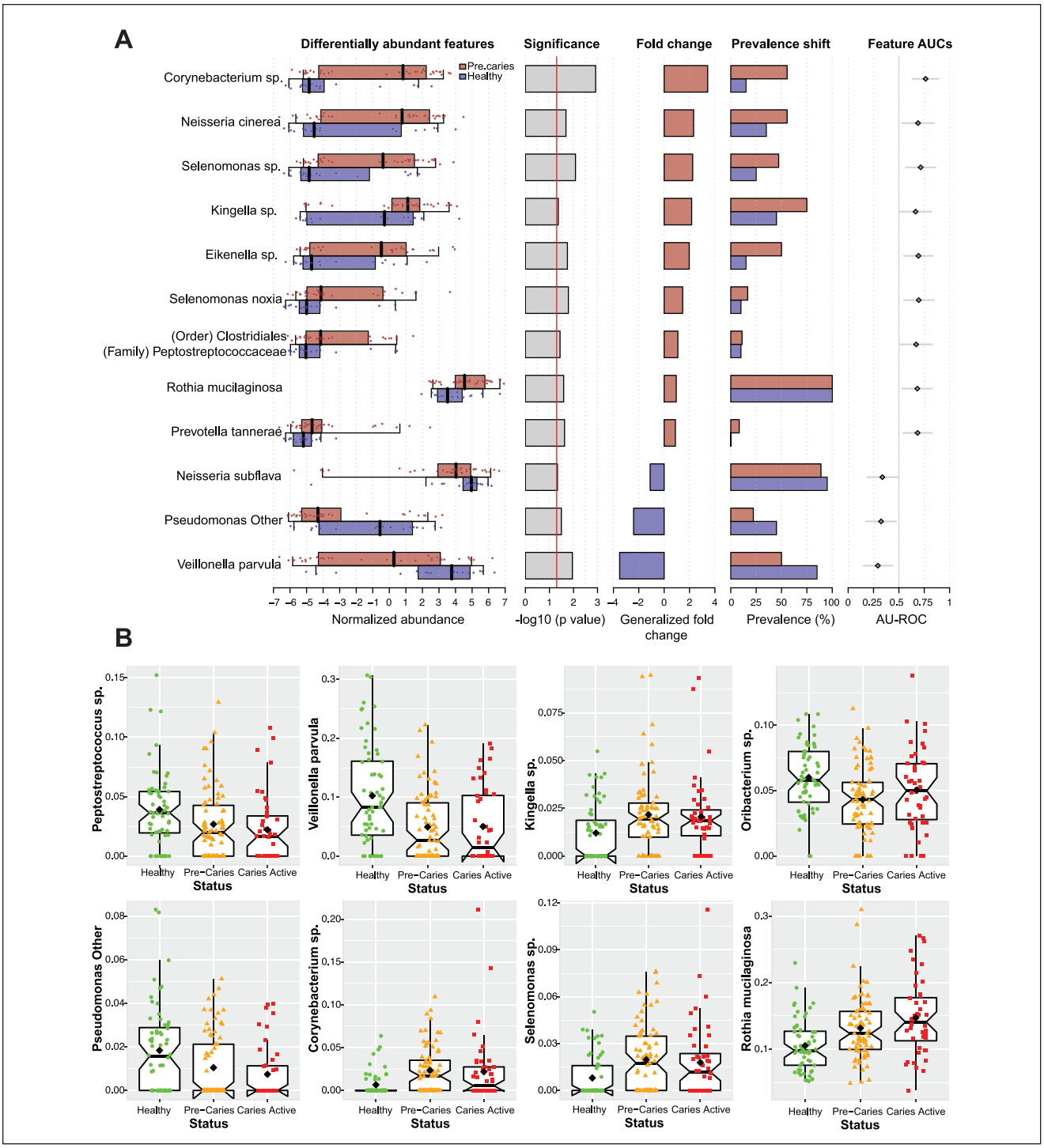


Figure 3. Differentially abundant microbes based on caries status. (**A**) Enrollment visit samples were tested for differential abundance of microbes between subjects who would go on to develop caries during the study period and those who would not using the Wilcoxon rank-sum test. After adjusting for multiple testing of all microbes, no significant differences were observed. Bacteria represented here (rows) are differentially abundant with an unadjusted *P* value of ≤0.05 and may indicate weak trends without being statistically robust on their own. The first column shows centered log ratio normalized abundances between groups. The second column shows the unadjusted significance of the difference between groups. The third and fourth columns shows the fold change between groups and the proportion of samples within each group that a microbe was detected in, respectively. The fifth column shows the area under the receiver operating characteristic curve (AU-ROC) associated with given bacteria as a nonparametric measure of enrichment. (**B**) Microbial differential abundance between all 3 caries status groups across all visits was tested using a multivariate mixed-effects model to control for age, sequencing reads, and repeated sampling of subjects. The normalized relative abundances of microbes differing significantly after multiple test correction are plotted on the *y*-axes and grouped by caries status on the *x*-axes. Boxplots are centered on the median, with notches indicating an approximately 95% confidence interval, boxes indicating the boundaries of the first and third quartiles, and whiskers extending to the largest and smallest values no further than 1.5 * (interquartile range) from the boxes. Points beyond the whiskers are outliers. If the notches of 2 boxes within the same panel do not overlap on the *y*-axis, there is strong evidence that the true medians differ. The black diamond indicates the mean of each group.

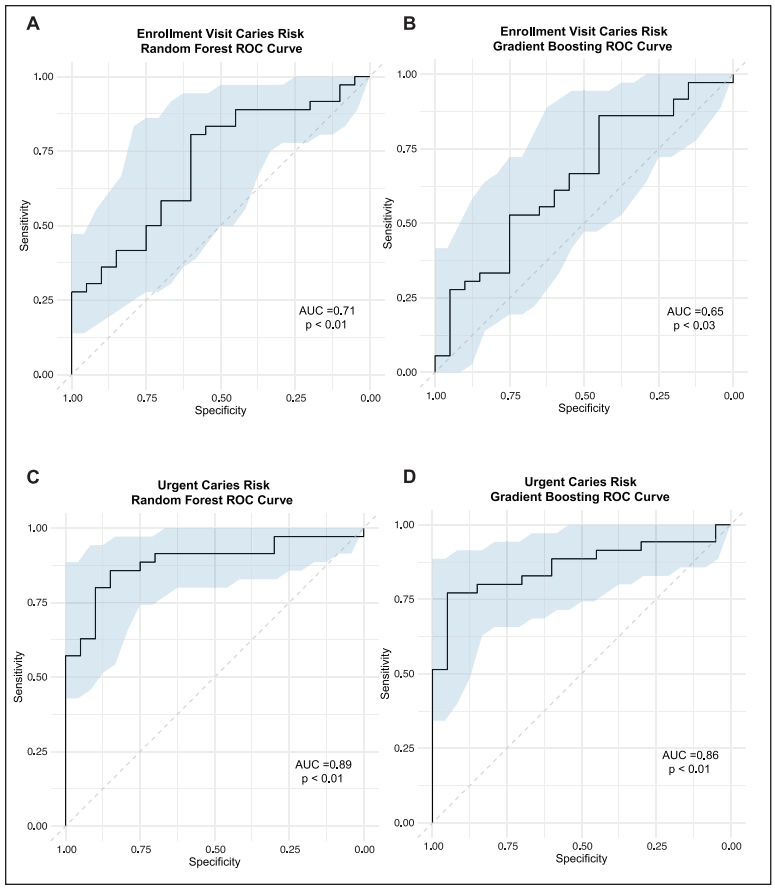


Figure 4. Caries risk classifier performance. Two types of machine learning classifiers, random forest models (**A**, **C**) and gradient tree boosting models (**B**, **D**), were trained and tested using nested cross-validation to predict caries risk based on saliva microbiota composition at enrollment (A, B) and at the visit immediately preceding caries development (or a corresponding caries-free visit in healthy control subjects) when caries risk is most urgent (C, D). Receiver operating characteristic (ROC) curves showing the sensitivity and selectivity of the models on held-out testing samples at various discrimination thresholds are plotted with 95% confidence intervals. The total area under the curve (AUC) is given as a metric of each model's predictive ability, and a *P* value is provided addressing the null hypothesis that the AUC is ≤0.5 (i.e., the model's predictive ability is no better than random chance). All models are able to discriminate between children who will develop caries within the following 6 to 24 mo and those who will not significantly better than chance, and both types of models are able to predict caries risk most accurately when children are within 6 mo of caries development.

selected as important, respectively, while the gradient boosting classifier had 17 and 14 (Fig. 5). Most of the important features were shared by both models within a given time point but not between time points, with the notable exceptions of *R. mucilaginosa, Streptococcus* sp., and *V. parvula*, which were important in all 4 models. In descending order, the top 5 most important features at the baseline time point were *R. mucilaginosa, Streptococcus* sp., *Corynebacterium* sp., *Neisseria subflava*, and *V. parvula* for the random forest and *R. mucilaginosa, Campylobacter* sp., *Prevotella* sp., *Streptococcus* sp., and *Veillonella dispar* for the gradient boosting classifier. At the immediate precaries time point, they were *Streptococcus* sp., *Rothia dentocariosa, R. mucilaginosa, Neisseria* sp., and an unknown member of the Gemellaceae family for the random forest and *R. mucilaginosa, Streptococcus* sp., *Capnocytophaga* sp., *Granulicatella* sp., and *R. dentocariosa* for the gradient boosting classifier. All of the taxa that were identified as significant by the initial multivariable test of differential abundance by caries status across all samples were selected as important in the classification models with the exception of *Oribacterium* sp. and *Peptostreptococcus* sp., the latter of which was only significantly different between healthy and caries-active states (not pre-caries) and was strongly associated with age.

Discussion

ECC is a chronic and destructive oral disease that afflicts vulnerable children often living in poverty and with limited access to dental care (Casamassimo et al. 2009). The disease is multifactorial, with development influenced by interactions between a cariogenic microbiota and host genetics, immunity, and diet (Simon-Soro and Mira 2015; Pitts et al. 2017; Bowen et al. 2018). Previous work has identified an association of ECC with *Streptococcus mutans, Lactobacillus, Viellonella*, and *Prevotella* along with an increase in ECC severity due to coinfection of *S. mutans* and *Candida albicans* (Xiao et al. 2018).

Although these cariogenic bacteria have been associated with ECC, they have not been demonstrated to be necessary or sufficient to cause the disease (Simon-Soro and Mira 2015). We undertook this study to identify oral microbiota taxa that are causal for ECC and to develop predictive models of ECC risk and prevention.

We first performed descriptive analyses of differentially abundant microbial taxa between persistently caries-free healthy controls and pre-caries plus caries-active states in initially caries-free preschool children. Five taxa—*Corynebacterium* sp., *Selenomonas* sp., *Kingella* sp., *V. parvula*, and unclassified *Pseudomonas*—were identified as differentially abundant between caries states by all 3 of our descriptive tests, suggesting that they are robust biomarkers of ECC risk and insensitive to the confounding effect of age and statistical procedures. Next, we developed machine learning models to predict ECC risk that are generalizable and sensitive up to 2 y prior to ECC onset and become increasingly selective as the risk of ECC onset becomes imminent, without losing sensitivity. Among all predictive models, 3 taxa were consistently important: *Streptococcus* sp., *R. mucilaginosa*, and *V. parvula*. Of these 3 taxa, the *Streptococcus* sp. and *V. parvula* have long been associated with dental caries (Simon-Soro and Mira 2015; Cross et al. 2016; Burne 2018). The dominant presence of *Streptococcus* sp. is likely due to the high levels of carbohydrates often found in the diets of less affluent families (Pitts et al. 2017). Glucose is the principle nutrient for *Streptococci*,

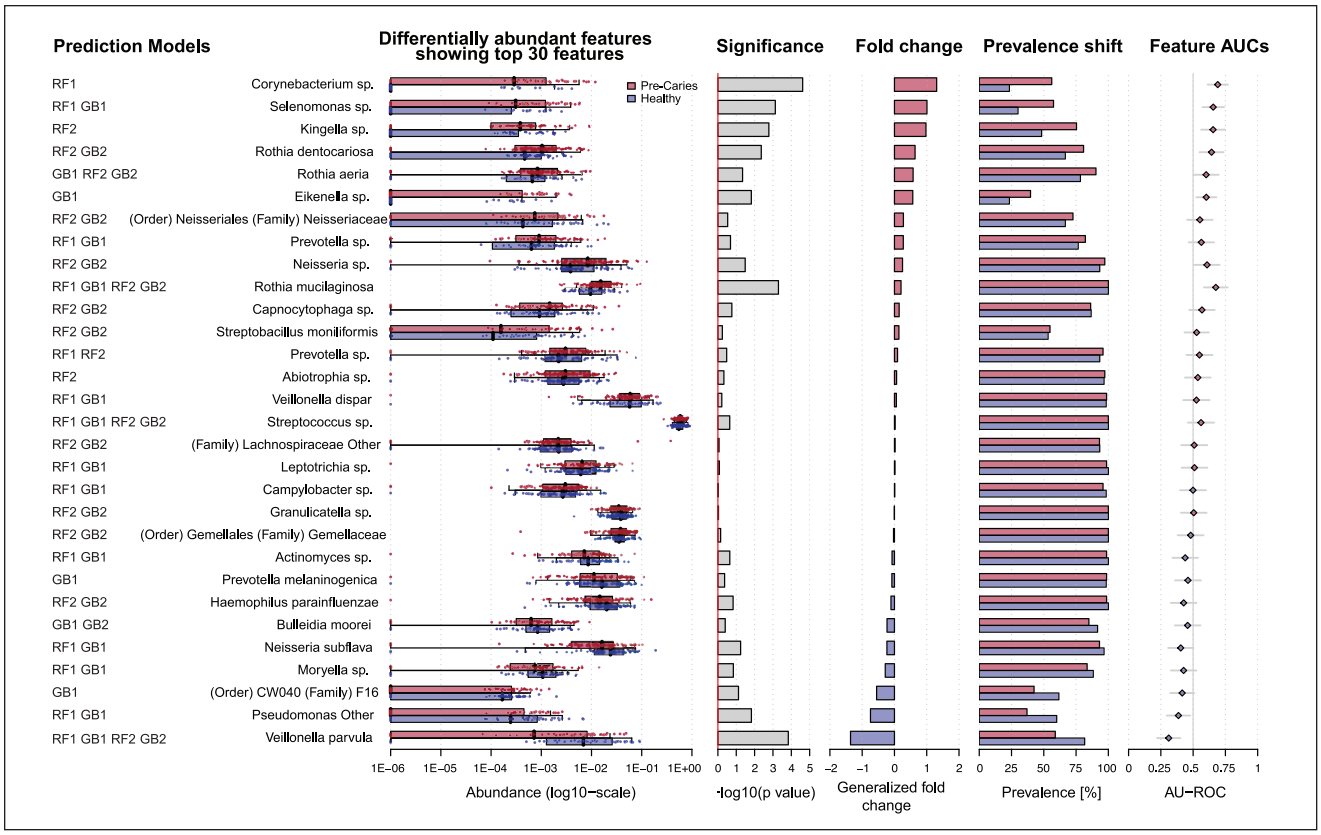


Figure 5. Important microbes in caries risk prediction models. For each time point (enrollment and immediately prior to caries development) and each type of model (random forest and gradient tree boosting), feature importance was averaged across all cross-validation folds and normalized to relative importance. For each model–time point combination, features were ranked by importance from greatest to least, and the top features with a cumulative relative importance of 50% were selected for plotting. The first column indicates which model or models a given microbe (row) was important in based on the selection criteria described: random forest at enrollment (RF1), random forest at urgent risk (RF2), gradient boosted at enrollment (GB1), and gradient boosted urgent risk (GB2). The second column indicates the most specific available taxonomic classification for the microbe. The following columns provide metrics of enrichment between pre-caries and healthy control groups across all time points. The final column gives the area under the receiver operating curve based on the rank abundance of the microbe and is independent of the machine learning models.

which rely on this carbohydrate for their carbon requirements and ability to extract energy from metabolism of sugars (Moye et al. 2014; Baker et al. 2017). *V. parvula* and other oral *Veillonella* sp. are early oral colonizers and metabolize the lactic acid produced by *Streptococci* to establish early multispecies biofilm communities on hard tooth surfaces (Periasamy and Kolenbrander 2010). *R. mucilaginosa* was identified in previous ECC studies as enriched in children with active caries (Dashper et al. 2019; Vieira et al. 2019) and has been identified as an important initiator of cell-cell interactions in early oral biofilms (Palmer et al. 2017). Together, the presence of these 3 robust biomarkers in the childhood oral cavity illustrates a mutualistic biofilm community, initiated by *V. parvula* and *R. mucilaginosa* but metabolically dependent on *S. mutans*–produced lactate to establish an ecological framework that promotes a caries-active state. These findings reveal persistent microbial biomarkers of cariogenic oral conditions, support the ecological plaque hypothesis, and, most important, demonstrate that oral microbiota signatures can be accurate, precise, and robust predictive indicators of ECC risk in young children who were initially caries-free. Given the 2-y prospective design, the findings provide supportive evidence of a causal link between oral microbiome and ECC in young children. We acknowledge that the small cohort used for our current analysis and the actual strength of the evidence are suggestive of but not compelling for causal associations. These limitations will be addressed in the future analysis of the entire 189-child cohort.

Nominally similar studies have been reported by Teng et al. (2015) and Xu et al. (2018). Both describe the use of machine learning models based on oral microbiota composition to assess caries risk in a longitudinally sampled cohort of initially caries-free preschool age children. However, important factors underpinning interpretation and applicability fundamentally distinguish them from this study. First, both of their models were informed by differences in oral microbiota composition between caries-free and caries-active states that could then be used to classify caries-free individuals with more "caries-like" microbiota composition as at risk for ECC. This difference is significant conceptually, in terms of model interpretation, because while pre-caries samples may become more caries-like than healthy caries-free samples, the difference between healthy caries-free and caries-active is not the same as the

difference between healthy caries-free and pre-caries states. Our models learn only these latter differences and incorporate no information about the composition of caries-active states. Second, their models were not tested for the ability to discriminate between pre-caries and caries-free healthy states but instead were each only tested on samples of a single class—either all precaries or all caries-free healthy—leaving their practical utility as predictors of ECC risk, which would depend on their receiver operating characteristics, an open question. Furthermore, both studies report a final model of caries risk tested on a single holdout data set, which does not provide a strong estimate of generalization performance. The nested cross-validation approach used in this study yielded much stronger evidence of good generalization performance.

The 56 subjects in this study is approximately the minimum number that is generally considered suitable for the kinds of machine learning models used (Vabalas et al. 2019). Therefore, a larger cohort would be desirable for improving model fit and necessary for validation of the results reported here. Furthermore, given that short-read 16S rRNA gene sequencing provides limited taxonomic resolution and no functional information, shotgun sequencing would likely prove to be a more powerful assay using an otherwise identical approach. Finally, a systems biology approach that incorporates the predictive power of microbiota-based CRA methods with conventional CRA variables was not explored in this study (Inquimbert et al. 2019). While this integrated approach should optimize recommendations for prevention of ECC, we find compelling evidence of the utility of the method employed here. In conclusion, saliva microbiota profiling coupled with machine learning represents a promising approach to accurate, reliable, robust CRA in initially caries-free preschool children.

Author Contributions

A. Grier, contributed to conception, design, data analysis, and interpretation, drafted and critically revised the manuscript; J.A. Myers, contributed to conception, design, data analysis, and interpretation, critically revised the manuscript; T.G. O'Connor, R.G. Quivey, contributed to conception, design, and data interpretation, critically revised the manuscript; S.R. Gill, contributed to conception, design, data acquisition, and interpretation, drafted and critically revised the manuscript; D.T. Kopycka-Kedzierawski, contributed to conception, design, and data interpretation, drafted and critically revised the manuscript. All authors gave final approval and agree to be accountable for all aspects of the work.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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